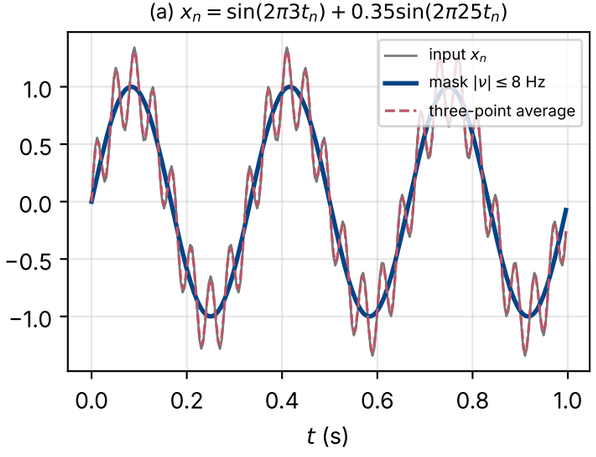
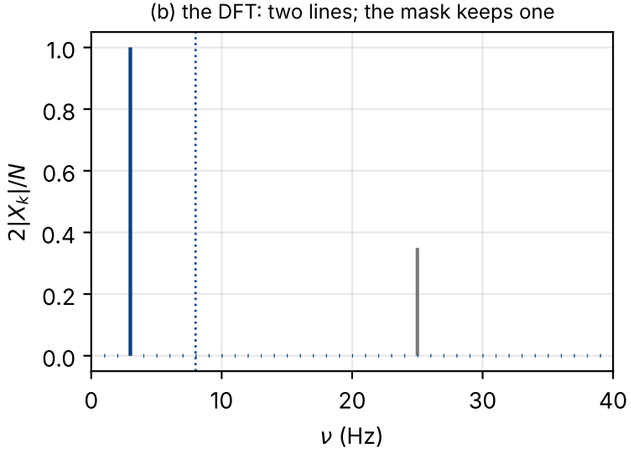
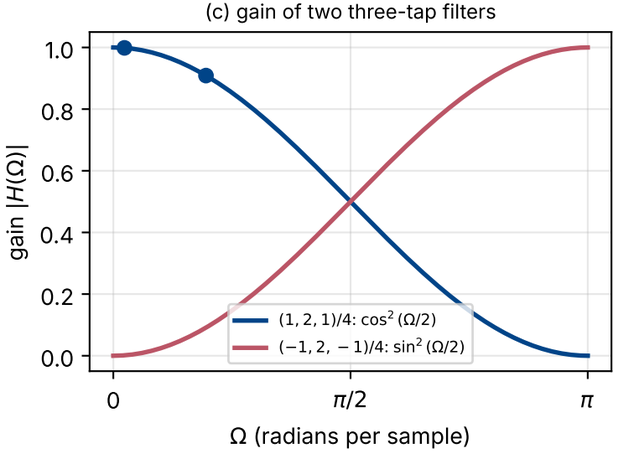
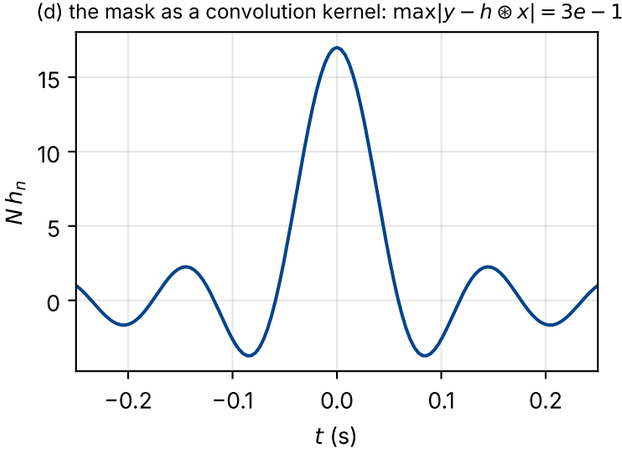

(a) $x_n = \sin(2\pi 3 t_n) + 0.35\sin(2\pi 25 t_n)$
input x_n
mask $|\nu| \le 8$ Hz
three-point average
1.0
0.5
0.0
−0.5
−1.0
0.0
0.2
0.4
0.6
0.8
1.0
t (s)


(b) the DFT: two lines; the mask keeps one
$2|X_k|/N$
1.0
0.8
0.6
0.4
0.2
0.0
0
10
20
30
40
ν (Hz)


(c) gain of two three-tap filters
gain $|H(\Omega)|$
1.0
0.8
0.6
0.4
0.2
0.0
$(1, 2, 1)/4$: $\cos^2(\Omega/2)$
$(-1, 2, -1)/4$: $\sin^2(\Omega/2)$
0
$\pi/2$
π
Ω (radians per sample)


(d) the mask as a convolution kernel: $\max|y - h \circledast x| = 3e-1$
$N\,h_n$
15
10
5
0
−0.2
−0.1
0.0
0.1
0.2
t (s)